

Onboarding Exercises - SURP 2026

Mylana Van Name

May 19, 2026

1 Exercise 1

1.1

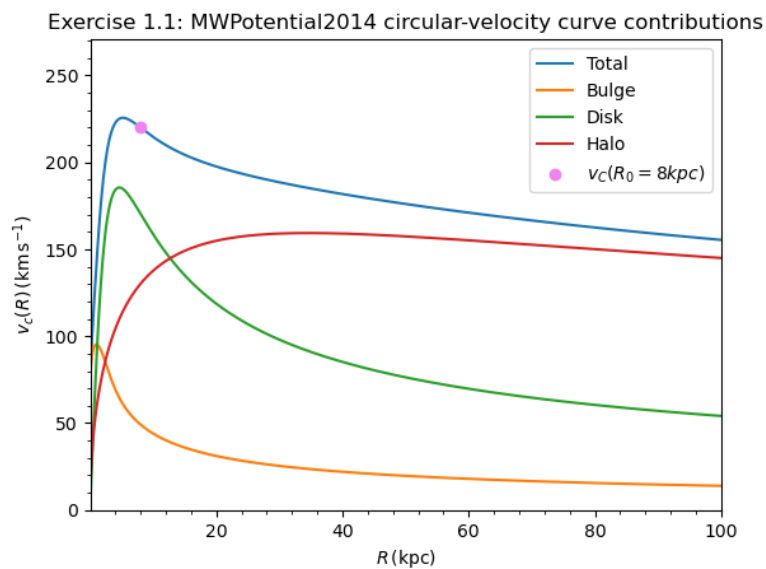


Figure 1.1: Exercise 1, question 1: MWPotential2014 circular velocity curve with bulge, disk, and halo components

1.2

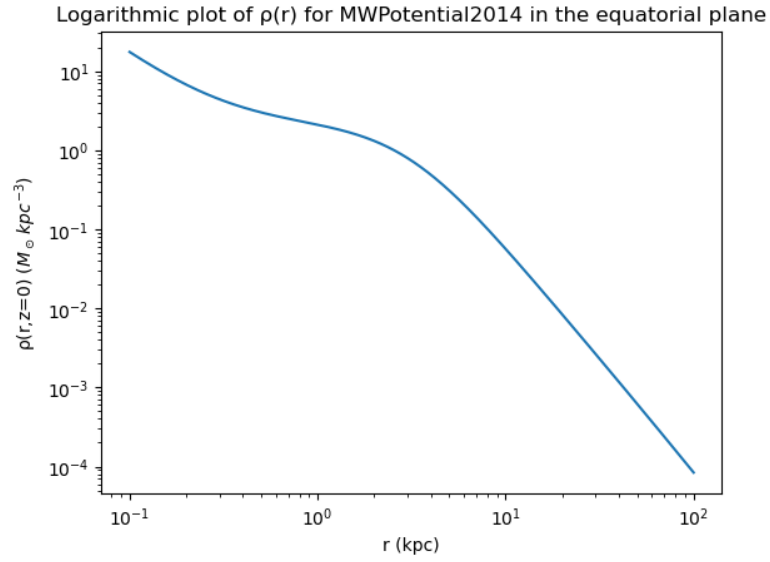


Figure 1.2: Exercise 1, question 2: $\rho(r)$ in the equatorial plane

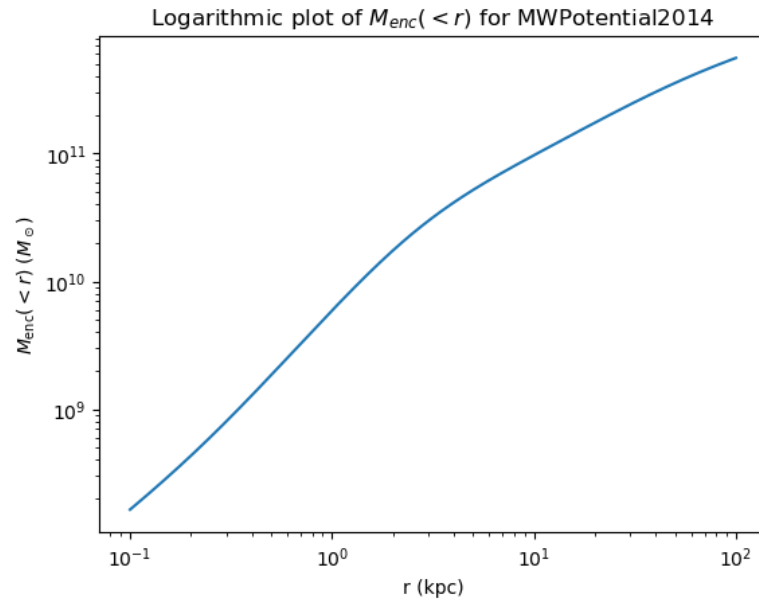


Figure 1.3: Exercise 1, question 2: $M_{enc}(< r)$

1.3

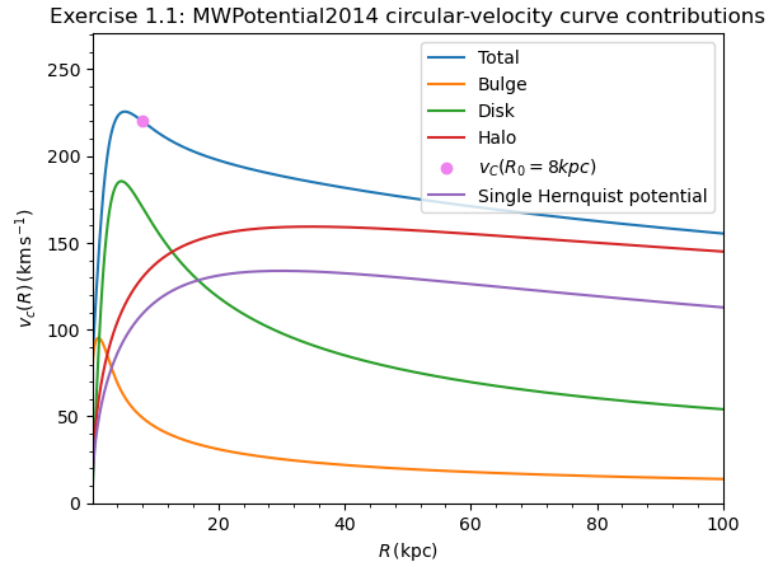


Figure 1.4: Exercise 1, question 3: Single Hernquist potential circular velocity compared with MWPotential2014 and its components

The single Hernquist potential's circular velocity curve matches the total Milky Way curve best at large R , although it is not a close match anywhere.

2 Exercise 2

2.2

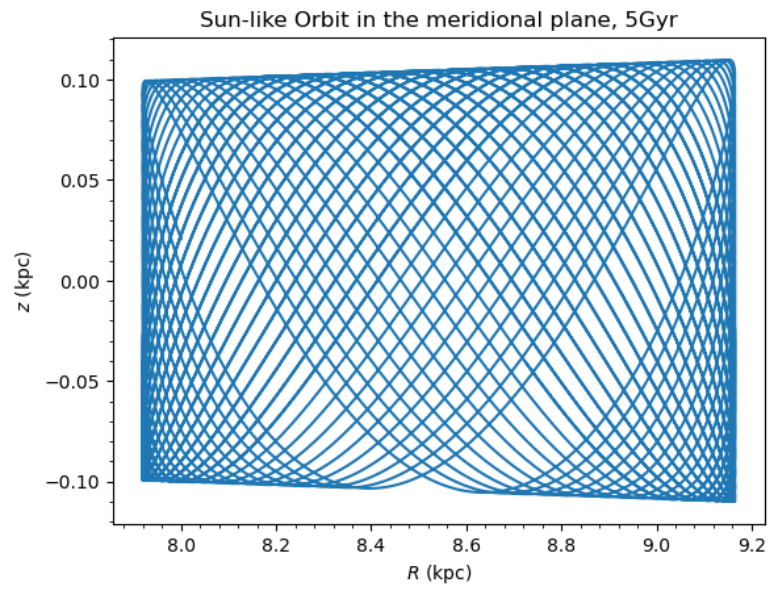


Figure 2.1: Exercise 2, question 2: Sun-like orbit in the (R, z) plane

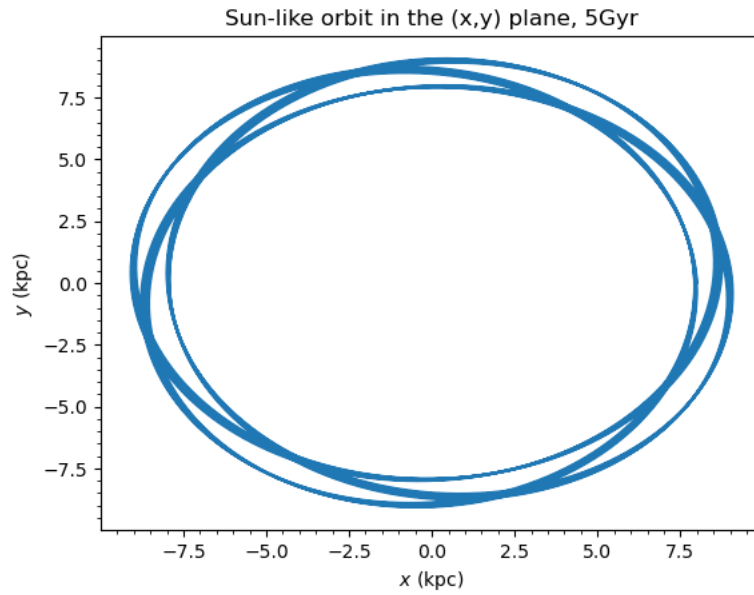


Figure 2.2: Exercise 2, question 2: Sun-like orbit in the (x,y) plane

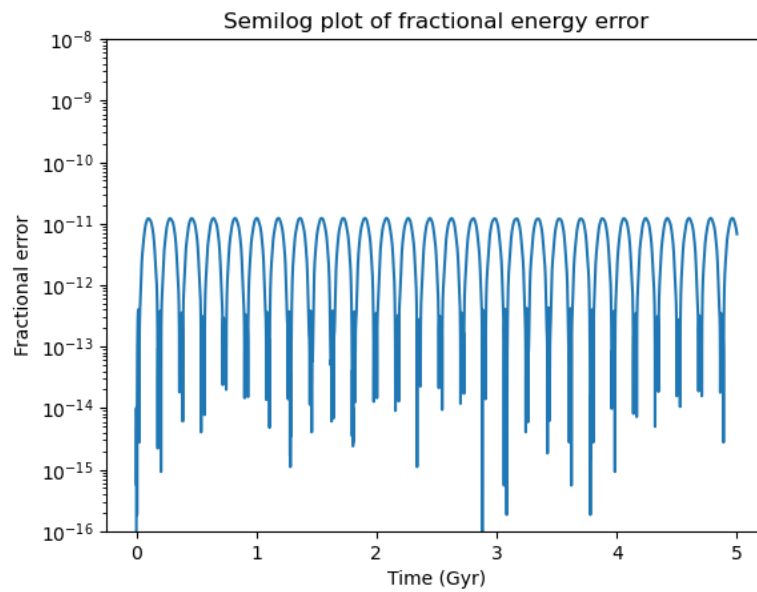


Figure 2.3: Exercise 2, question 2: Fractional energy error vs. time, 10000 step time grid, default integrator

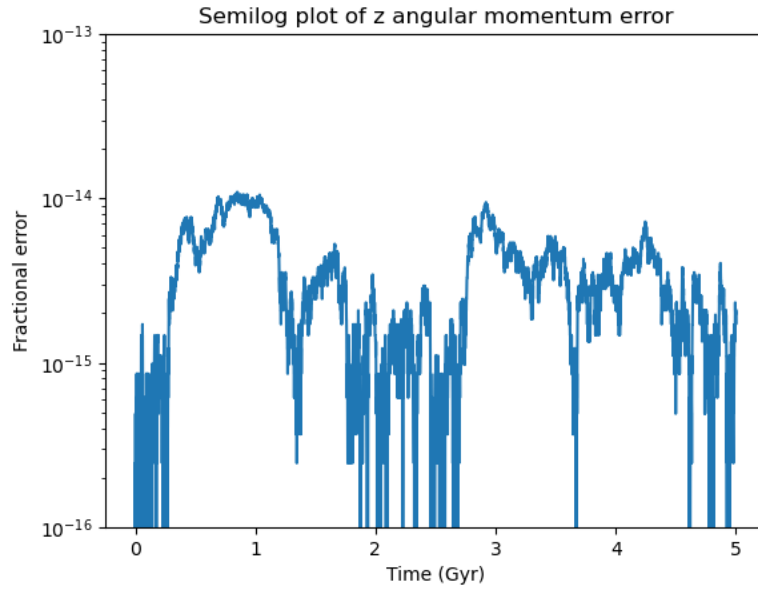


Figure 2.4: Exercise 2, question 2: Fractional z-angular-momentum error vs. time, 10000 step time grid, default integrator

Figures 2.3 and 2.4 show that the default integrator (symplec4_c) has a maximum fractional error of around 10^{-11} for energy and around 10^{-14} for z-angular-momentum (almost machine precision). These errors also appear to remain relatively stable over time.

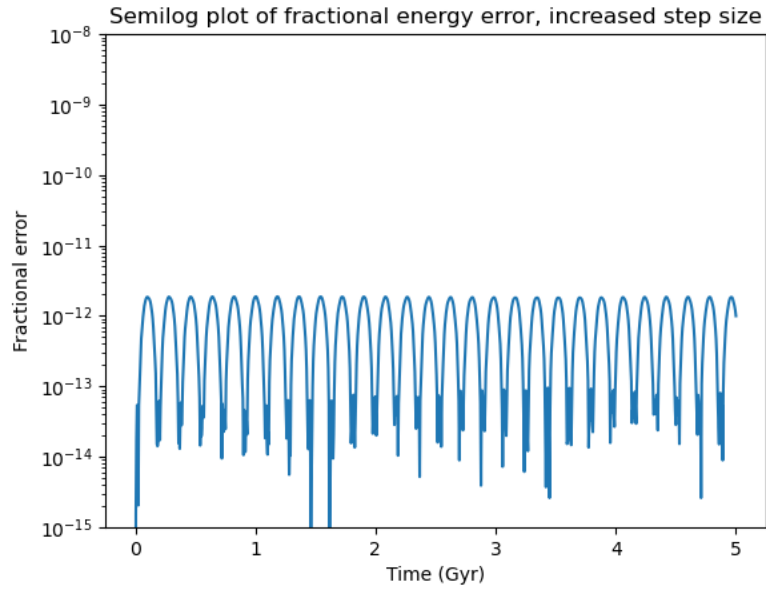


Figure 2.5: Exercise 2, question 2: Fractional energy error vs. time, 1000 step time grid, default integrator

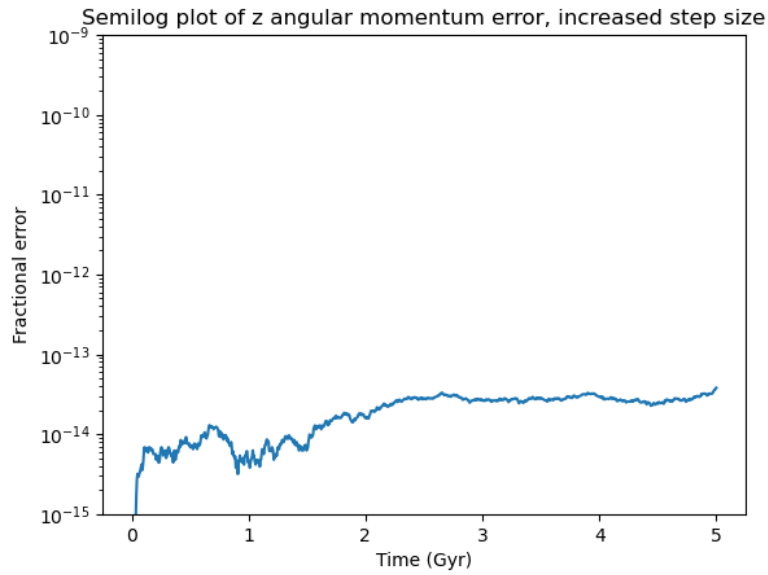


Figure 2.6: Exercise 2, question 2: Fractional z-angular-momentum error vs. time, 1000 step time grid, default integrator

Decreasing the number of steps from 10000 to 1000 did not meaningfully change the magnitude of fractional error for energy or z-angular-momentum for the default symplec4_c integrator.

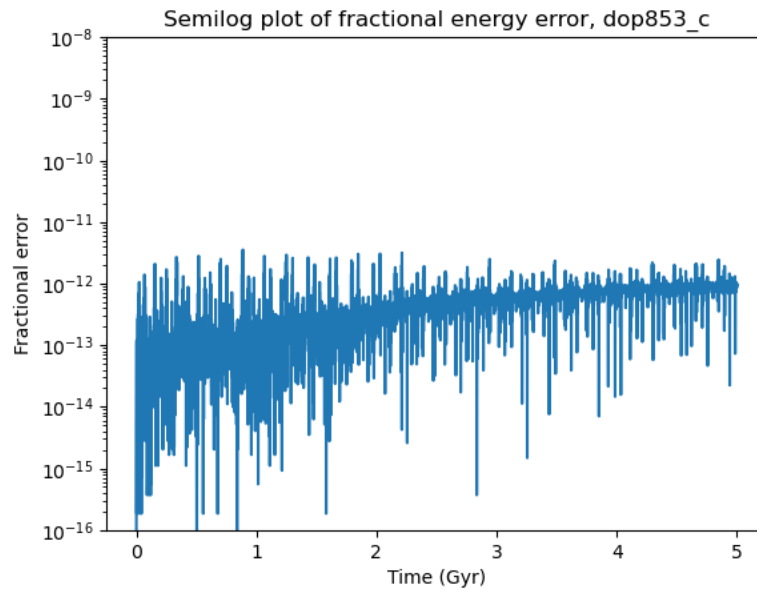


Figure 2.7: Exercise 2, question 2: Fractional energy error vs. time, 10000 step time grid, dop853_c integrator

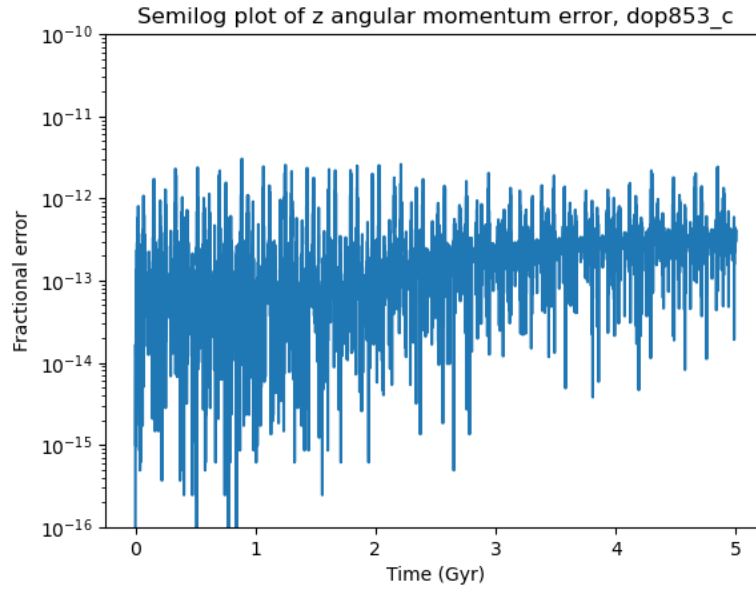


Figure 2.8: Exercise 2, question 2: Fractional z-angular-momentum error vs. time, 10000 step time grid, dop853_c integrator

The maximum fractional energy and z-angular-momentum errors for the dop853.c integrator appear slightly higher than for the default integrator.

2.3

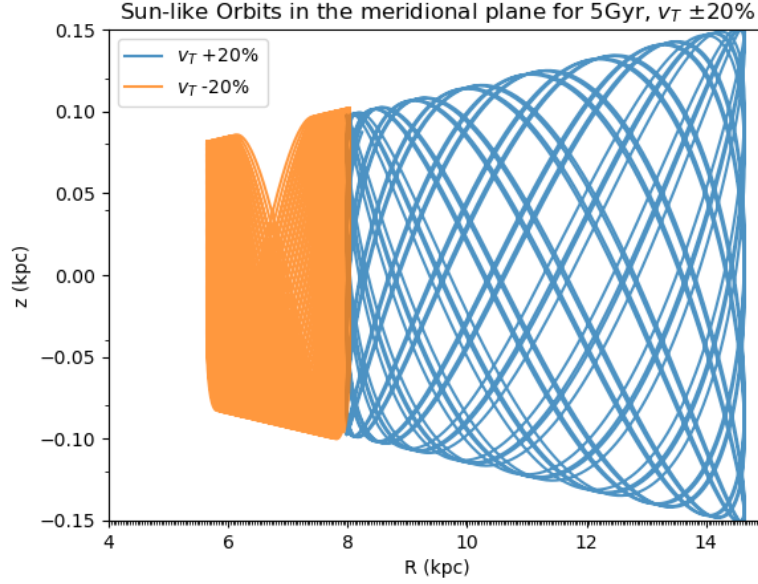


Figure 2.9: Exercise 2, question 3: v_T variation for Sun-like orbits in the meridional plane

Increasing the initial v_T by 20% produced a more radial orbit than lowering it by 20%, since Figure 2.9 shows that higher- v_T orbit has a much larger range of R values than the lower- v_T orbit.

3 Exercise 3

3.2

The result for the best-fit (M, a) for the single Hernquist potential fit was $(1.0075e12 M_{sun}, 7.9348 kpc)$.

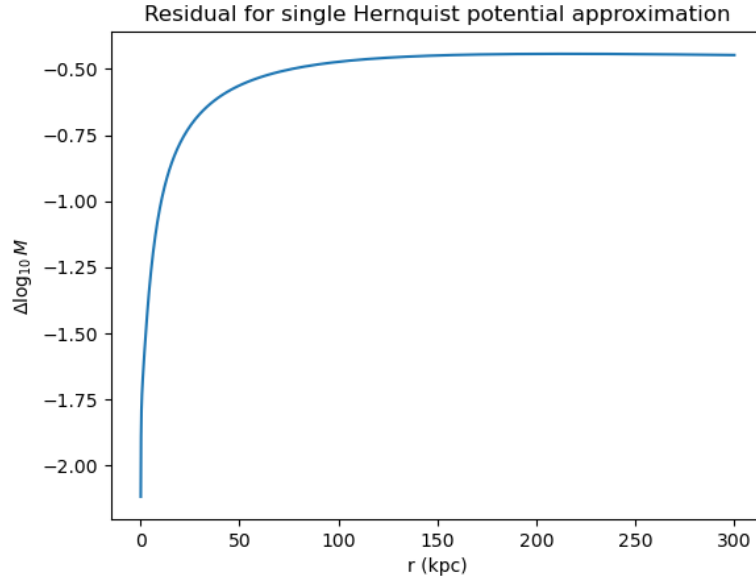


Figure 3.1: Exercise 3, question 2: Residual plot for single Hernquist approximation

3.3

For the two-component fit, the best-fit (M, a) for the Hernquist (bulge) component was $(1.1676e11 M_{sun}, 2.3180 kpc)$. For the NFW (halo) component, the best-fit (M, a) was $(3.7103e11 M_{sun}, 11.7741 kpc)$.

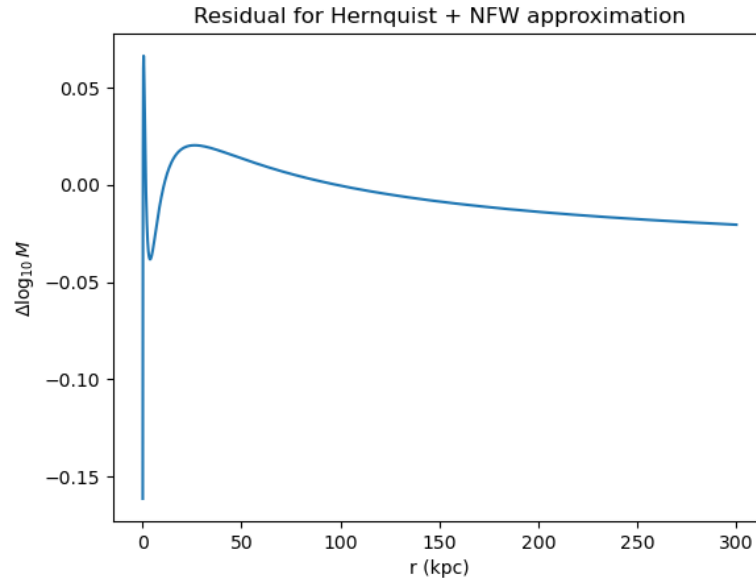


Figure 3.2: Exercise 3, question 3: Residual plot for Hernquist + NFW approximation

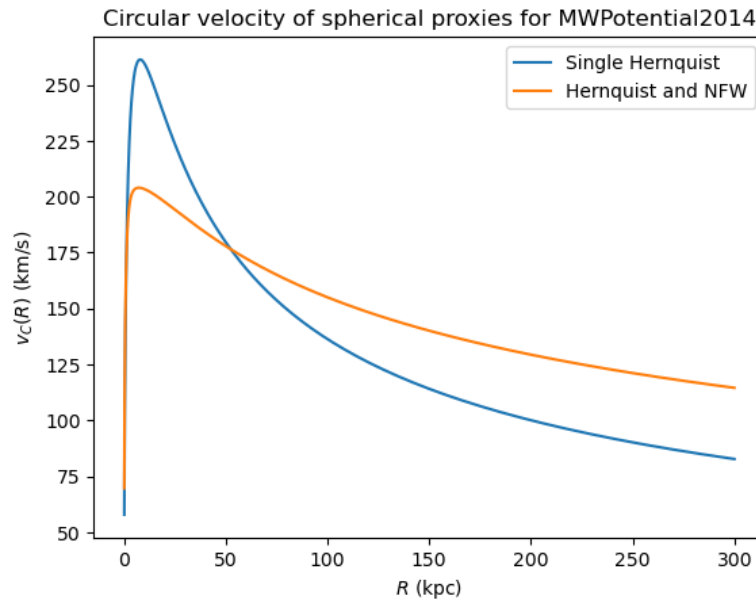


Figure 3.3: Exercise 3, question 3: $v_c(R)$ for single Hernquist and two-component fits for the Milky Way

3.4

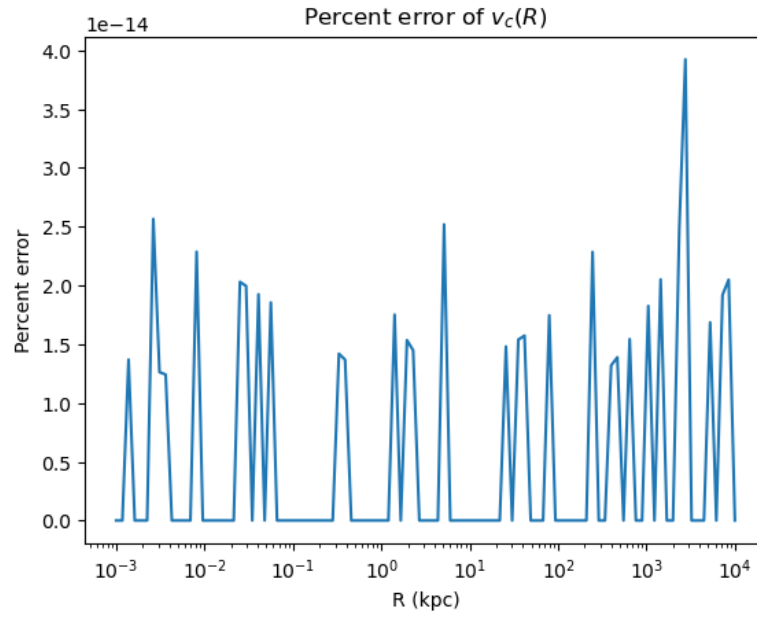


Figure 3.4: Exercise 3, question 4: Percent error in $v_c(R)$ for spherical Milky Way proxy

4 Exercise 4

4.2

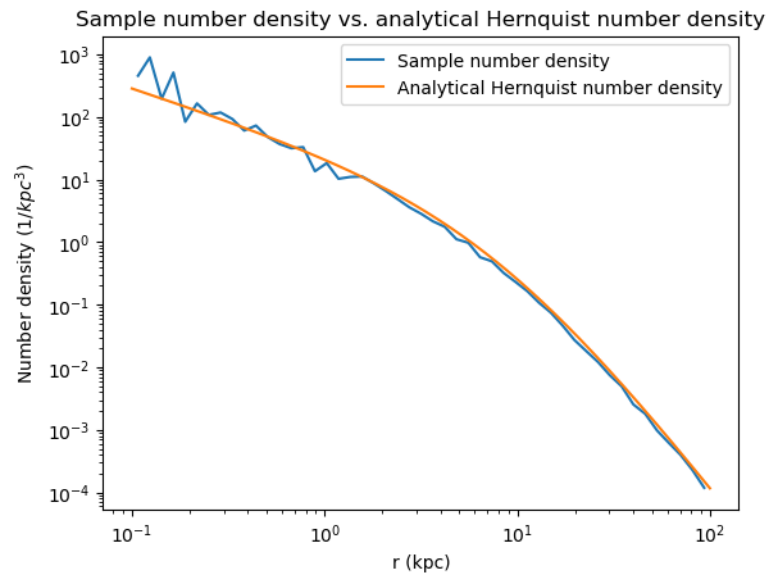


Figure 4.1: Exercise 4, question 2: Radial number density profile for sample vs. analytic Hernquist

4.3

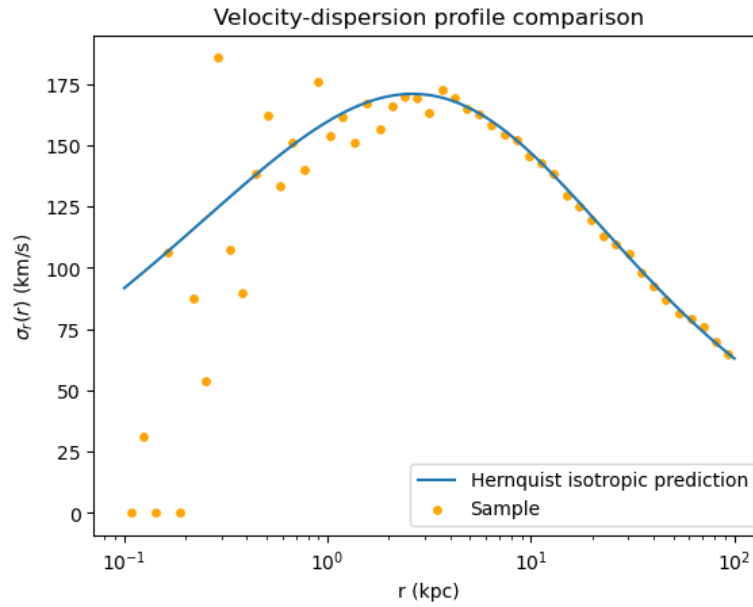


Figure 4.2: Exercise 4, question 3: Comparison between sample velocity dispersion profile and analytic Hernquist isotropic prediction

The sample velocity dispersion matches the Hernquist prediction very well at larger r , although it lacks accuracy at small r ($< 1 \text{ kpc}$).

4.4

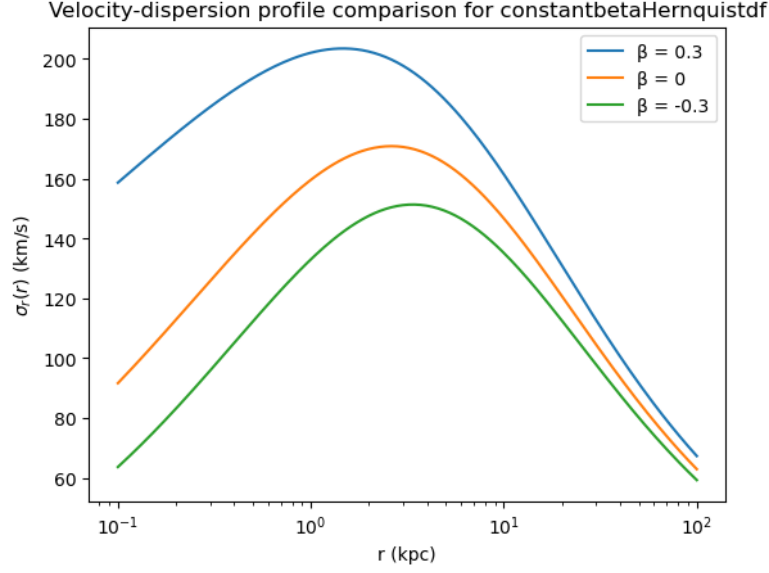


Figure 4.3: Exercise 4, question 4: Change in velocity dispersion profile for $\beta = -0.3$ and $\beta = +0.3$

The positive β value increased the velocity dispersion profile, which signifies more radial dispersion, while the positive β value decreased the velocity dispersion profile, signifying more tangential dispersion.

5 Exercise 5

5.2

On average, using all available cores was around 3.5 times faster than using a single thread.

5.3

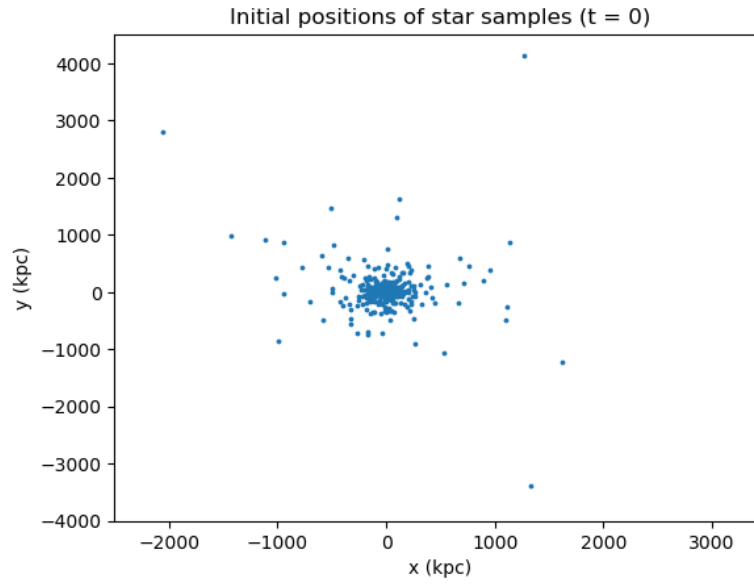


Figure 5.1: Exercise 5, question 3: Initial (x, y) projection of $N=2000$ star sample from isotropic Hernquist DF

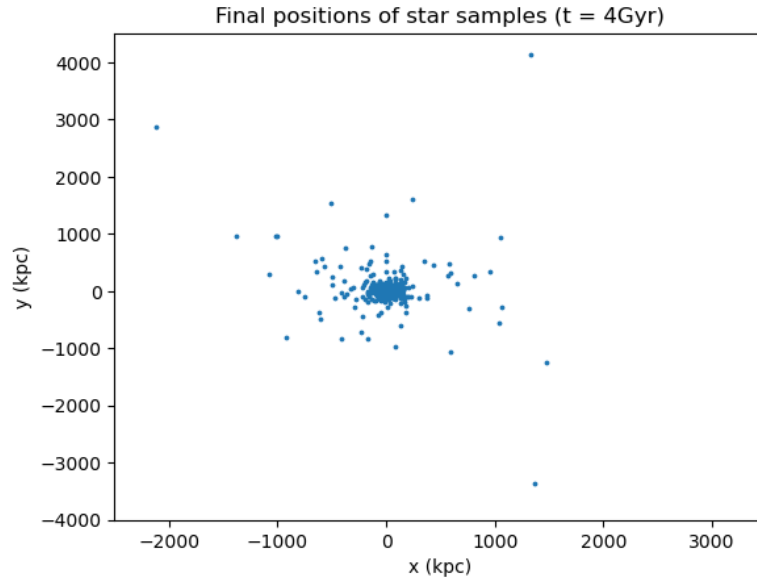


Figure 5.2: Exercise 5, question 3: Final (x, y) projection of $N=2000$ star sample from isotropic Hernquist DF

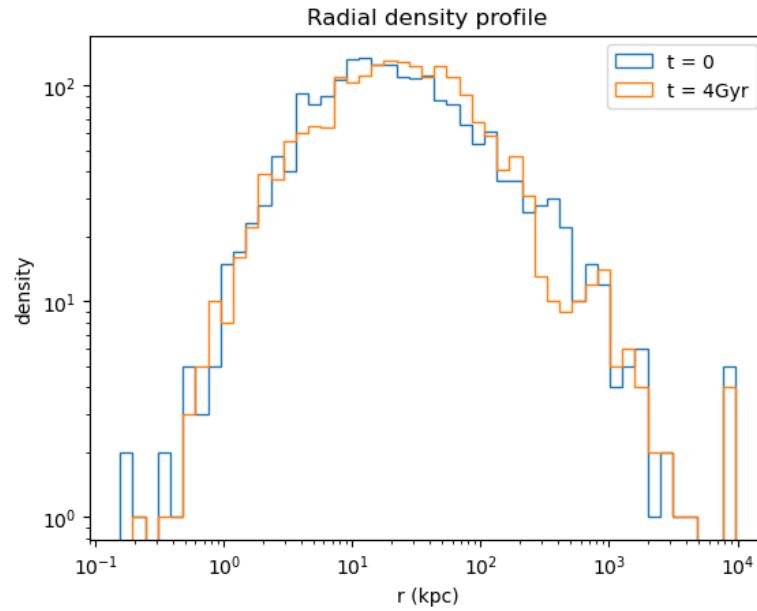


Figure 5.3: Exercise 5, question 3: Radial density profile at $t = 0$ and $t = 4$ Gyr

6 Exercise 6

6.2

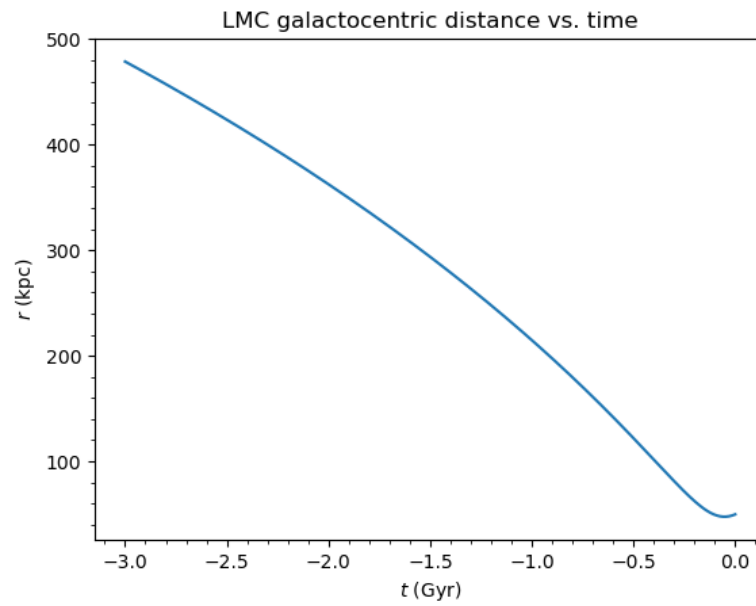


Figure 6.1: Exercise 6, question 2: Galactrocentric distance $r_{LMC}(t)$

6.4

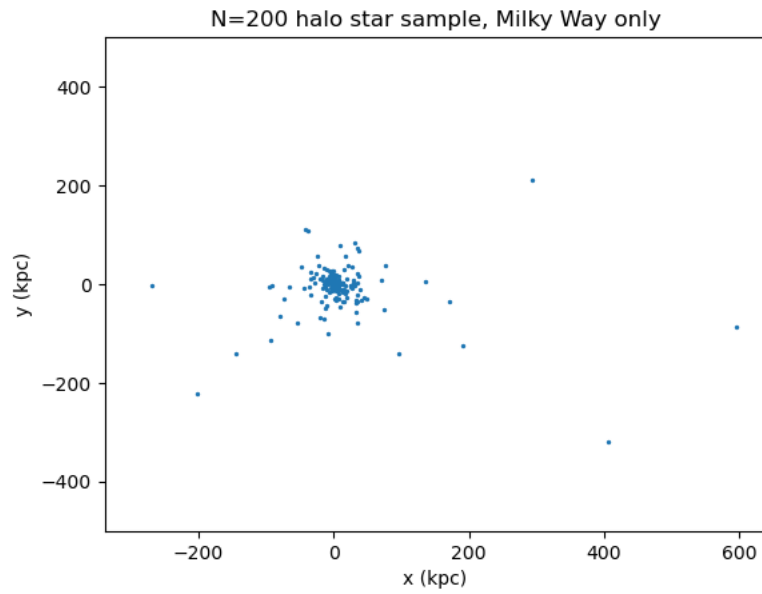


Figure 6.2: Exercise 6, question 4: Unperturbed $N=200$ halo star sample (Milky Way only)

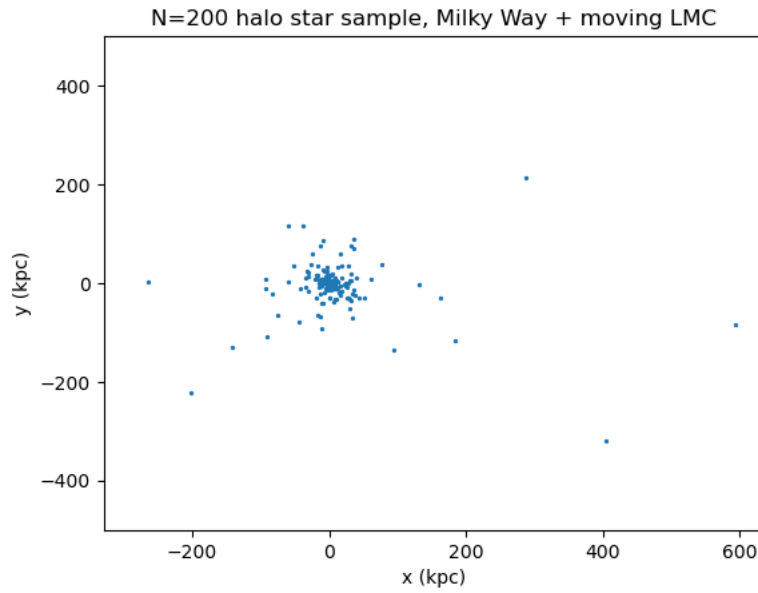


Figure 6.3: Exercise 6, question 4: $N=200$ halo star sample with Milky Way + moving LMC

Figure 6.3 shows that the halo's outer stars (larger r) were the most affected by the moving LMC perturbation.

7 Exercise 7

7.1

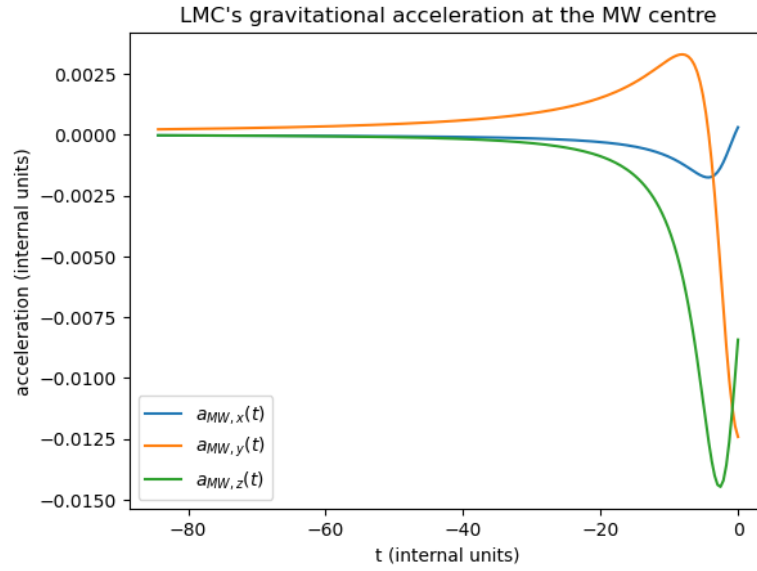


Figure 7.1: Exercise 7, question 1: Acceleration of the Milky Way due to the LMC vs. time

7.3

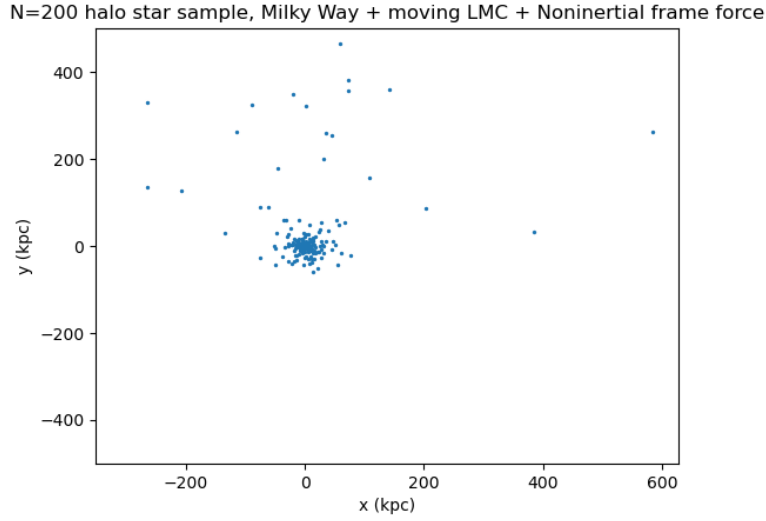


Figure 7.2: Exercise 7, question 3: N=200 Halo integration with Non-inertial frame force included

The non-inertial frame plot (Figure 7.2) shows stars scattered in the $+y$ direction in contrast with the dipole pattern observed in the plot without the non-inertial force included (Figure 6.3)

7.4

The LMC's mass pulls the centre of the Milky Way towards it, but the outer halo stars not as strongly bound to the centre and are therefore not accelerated as strongly towards the LMC. This causes the dipole pattern where there is an overdensity of outer halo stars in the $+y$ direction, implying that the LMC is roughly in the $-y$ direction. This effect is missed if the non-inertial frame is not taken into account. Since the LMC's mass is what causes the Milky Way centre to accelerate, the assumed LMC mass is directly related to the magnitude of the acceleration and therefore a more massive LMC increases the amplitude of the reflex term.